



College of Computing Education

Bachelor of Science in Information Technology

Employee Management System for MatinaTech IT Solutions Inc.

IT6L (8419) Fundamentals of Database Systems

Submitted by:
Bolongaita, Eliza Marie

Submitted to:
Sheila Marie Villarin

April 30, 2026

TABLE OF CONTENTS

III. INTRODUCTION	3
Background of the System	3
Description of Current Process	3
Importance of the Project	4
IX. PROBLEM STATEMENT	4
V. PROPOSED SOLUTION	5
VI. OBJECTIVES OF THE STUDY	5
General Objective	5
Specific Objective	5
VII. SCOPE AND LIMITATIONS	6
Scope	6
Limitations	6
VIII. SYSTEM ANALYSIS AND DESIGN	7
Business Rules	7
Conceptual Design	8
Entity Mapping	8
Relationships	8
Processes	9
IX. DATABASE DESIGN	9
List of Tables	9
Fields and Datatypes	9
Constraints	9
X. ADVANCED DATA FEATURES	9
Triggers	9
Views	10
XI. SYSTEM FEATURES AND FUNCTIONALITIES	10
System Modules	10
Features	10
XII. PROTOTYPE / USER INTERFACE	10
Login Page	10
Dashboard	10
Main Modules	10
Employee	10
Department	10
Attendance	10
Leave Request	10
XIII. TESTING AND RESULTS	10
XV. REFERENCES	11

III. INTRODUCTION

Background of the System

Employee management is an important part of any organization, as it plays a significant role in overall performance and success. In many companies such as MatinaTech IT Solutions Inc., one of the essential things that ensures a smooth HR operations is to have an accurate management of employee information, and accurate record keeping. Organizations invest heavily in managing their workforce since human resources are responsible for maintaining records, evaluating performance, and supporting decision-making processes [1][2]. However, managing employee data manually or through separate systems is time-consuming and often leads to inefficiencies, data inconsistencies, and difficulty in accessing information [3]. Studies have shown that implementing employee management systems improves organizational performance and productivity by enhancing structure, communication, and overall efficiency [4][5]. Additionally, such systems simplify administrative tasks, centralize data, and reduce errors, leading to improved operational effectiveness [6].

Description of Current Process

At MatinaTech IT solution Inc., employee-related processes, such as maintaining records, tracking attendance, and managing leave, are being handled manually or through separate systems. Manual management involves the use of non-technological tools such as paper, writing materials, and physical filing in the cabinets, while some organizations rely only on basic software such as Microsoft Word and Excel to store employee information. However, these methods are insufficient to organize data in a systematic and efficient manner. Manual filling systems occupy a big physical space and can lack proper organization and make the date retrieval time-consuming as records need to be searched individually. And there are instances where records can be misplaced, leading to data inconsistencies in data entry and updates, while the absence of a backup system poses risks to data security and potential loss of important information.

Importance of the Project

In addressing these challenges, an employee management system (HRIS) is proposed for MatinaTech IT Solution Inc. This is a web-based solution that is designed to centralize employee information and HR processes. This system is a role-based system wherein the system allows admins, HR staff, and managers to manage employee records and assign departments and monitor attendance that includes late entries and absences. It also handles rejecting and approving leave requests by the employees efficiently. This system also provides reporting features such as employee directories, attendance summaries, and leave reports. By automating these processes, the system improves efficiency and reduces errors and provides real-time access to information and ultimately enhances organizational productivity.

IX. PROBLEM STATEMENT

The current employee management process at MatinaTech IT Solution Inc., relies on manual methods and the use of separate tools such as paper-based records, Microsoft Word, and Microsoft Excel, which leads to several issues. These include inefficient data organization, time-consuming retrieval of employee records, and increased risk of human error in data entry and updates. Additionally, the manual filing system requires large physical storage space and may result in misplaced or lost records. The absence of a centralized system also makes it difficult to monitor attendance, track late entries, and manage leave requests effectively. Furthermore, the lack of data backup and security mechanisms increases the risk of losing important and confidential information [6].

V. PROPOSED SOLUTION

To address these issues, a web-based Employee Management System (HRIS) is proposed to MatinaTech IT Solution Inc., to provide a centralized and automated platform for handling employee information and HR processes. The system enables Admin, HR staff, and managers to efficiently manage employee records, assign departments, monitor attendance including late entries, and process leave requests within a single system. It also includes reporting features for better tracking and decision-making. By replacing manual processes with a structured digital system, the proposed solution improves data accuracy, reduces processing time, enhances data

security through proper storage and backup, and provides real-time access to information, resulting in a more efficient and reliable employee management process.

VI. OBJECTIVES OF THE STUDY

General Objective

To develop a web-based application, an employee management system (HRIS) for MatinaTech IT Solution Inc., using the Laravel framework that efficiently manages the employee information, attendance records, leave requests, and department assignments and information in a centralized and organized system.

Specific Objective

- To manage employee information, profiles, and employment details by providing a system that allows the creation, updating, and storage of employee personal information.
- To track an employee's attendance and work status through recording employee attendance time in and out, as well as monitoring the attendance status of each record, such as late, present, absent, or on leave.
- To process leave requests more efficiently and approvals by enabling employees to submit leave request applications, allowing managers to review and approve and reject the application, and HR to review the request.
- To organize employees by department and position by assigning employees to their respective departments and roles for better workforce management structure.
- To generate a summary report for HR monitoring by providing automated reports such as employee lists, attendance summaries, and leave records for more efficient decision-making.

VII. SCOPE AND LIMITATIONS

In this section it defines the functionalities that are included in the system as well as the features that are not part of the system.

Scope

This system includes the following functionalities and features:

- User authentication and authorization that allows users to securely log in and access the system based on their assigned roles.
- Employee management enables the creation, updating, and viewing and also deletion of the employee records in the system.
- Department management, which allows organization and assignment of employees to a specific recorded department in the system.
- Attendance tracking, which allows the employee to record their attendance time in and out and monitor their attendance status, ensuring correct recording of attendance.
- Dashboard and reporting that provides an overview of system data and generates reports for HR monitoring
- A role-based access control that restricts system access based on their user roles, such as admin, HR staff, manager, and employee for more role-based applications.
- Search and filtering functionality that allows users to easily find and track employees and department attendance records, leave records based on specific date, name, department.
- User profile management that allows users in any roles to update their personal information and account details.

Limitations

The system does not include the following functionalities:

- Payroll processing, which is salary computation and financial transaction, is not part of the system.
- Biometric device integration in attendance recording, as the recording of attendance is done manually within the system.
- Recruitment and applicant tracking that are outside the scope of this system.
- Advanced performance analytics, such as detailed employee performance evaluation and predictive analysis.

VIII. SYSTEM ANALYSIS AND DESIGN

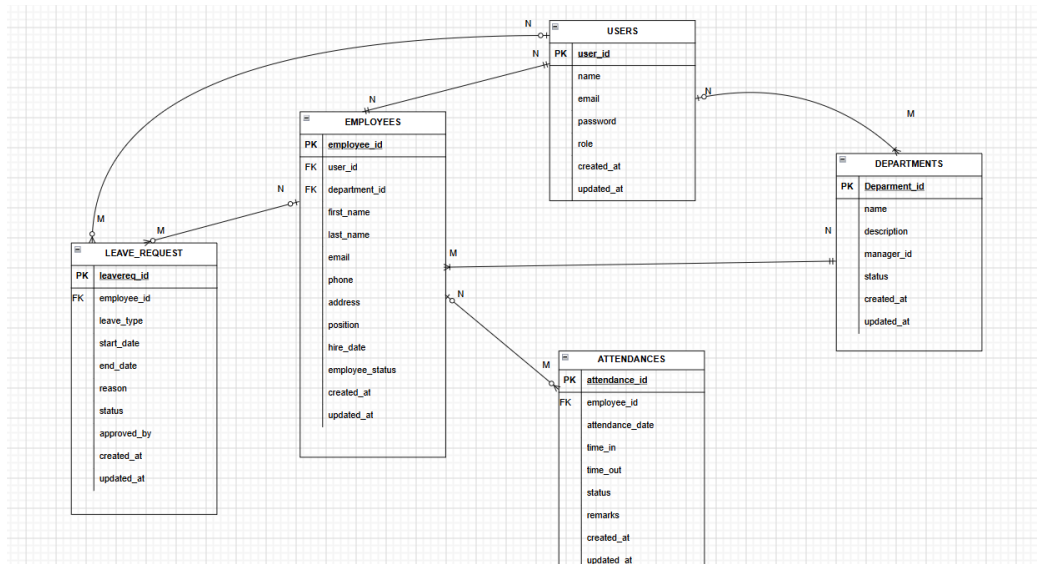
This section describes the business rules and the conceptual design and the explanation of the entities, relationships and the process that are followed and implemented in this project.

Business Rules

This section defines the guidelines and constraints that the system follows to ensure proper operation, data integrity, security, and compliance with user roles.

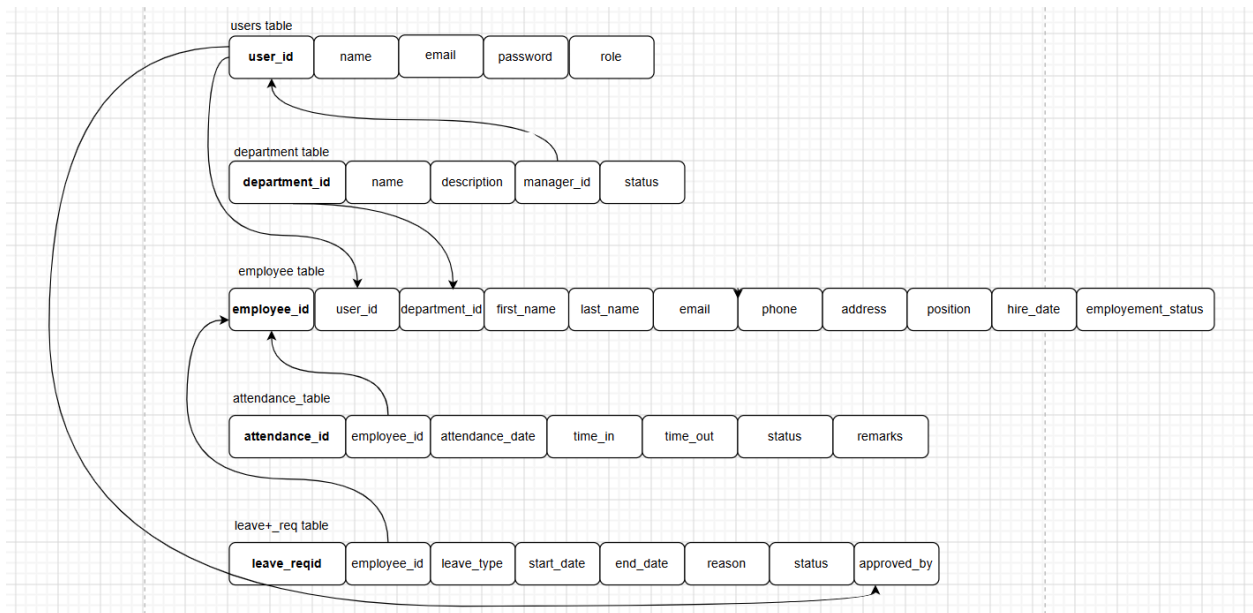
- Only authenticated users can access the system
- Only admins can manage user accounts
- Every employee must belong to one department
- Attendance entries must be linked to a valid employee
- A leave request must have a valid start date and end date
- Approved leave requests can only be changed by authorized users
- Reports must only display data allowed by user role
- An employee profile email must be unique

Conceptual Design



Explanation of entities, relationships, and processes

Entity Mapping



Relationships

- One department can have many employees
- One employee belongs to one department
- One employee can have many attendance records
- One employee can submit many leave requests
- One manager can approve many leave requests
- One user account may be linked to one employee profile

Processes

The system manages the employee-related operations such as

- The HR and admin will add an employee to the system, registering their personal details and assigning them to one department and granting the employee log-in credentials.
- The HR and admin will add a department to the system, registering its details from its ID to description and status and also assigning a manager and granting it full access to the employee records and employee list under their department.
- The employee that is registered can record their attendance by performing time-in and time-out, which is automatically stored and monitored by the system.

- Employees can also submit leave requests by entering their necessary information, which are sent to the manager to review.
- The manager evaluates the request and can either approve or reject it, and the system updates the leave request status.
- The system can generate reports such as employee lists, attendance summaries, and leave reports, which can be accessed by the authorized users and can be used for monitoring and decision-making purposes only.

IX. DATABASE DESIGN

List of Tables

☒	attendances	★	Browse	Structure	Search	Insert	Empty	Drop	5	InnoDB	utf8mb4_unicode_ci	32.0	KiB
☐	cache	★	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0	KiB
☐	cache_locks	★	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0	KiB
☒	departments	★	Browse	Structure	Search	Insert	Empty	Drop	4	InnoDB	utf8mb4_unicode_ci	32.0	KiB
☒	employees	★	Browse	Structure	Search	Insert	Empty	Drop	6	InnoDB	utf8mb4_unicode_ci	64.0	KiB
☐	failed_jobs	★	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0	KiB
☐	jobs	★	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0	KiB
☐	job_batches	★	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	16.0	KiB
☒	leave_requests	★	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	48.0	KiB
☐	migrations	★	Browse	Structure	Search	Insert	Empty	Drop	10	InnoDB	utf8mb4_unicode_ci	16.0	KiB
☐	password_reset_tokens	★	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	16.0	KiB
☐	sessions	★	Browse	Structure	Search	Insert	Empty	Drop	4	InnoDB	utf8mb4_unicode_ci	48.0	KiB
☒	users	★	Browse	Structure	Search	Insert	Empty	Drop	8	InnoDB	utf8mb4_unicode_ci	32.0	KiB

Fields and Datatypes, Constraints

Users

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐ 2	name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 3	email 🔑	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 4	email_verified_at	timestamp			Yes	NULL			Change Drop More
☐ 5	password	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 6	role	enum('admin', 'hr', 'manager', 'employee')	utf8mb4_unicode_ci		No	employee			Change Drop More
☐ 7	remember_token	varchar(100)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 8	created_at	timestamp			Yes	NULL			Change Drop More
☐ 9	updated_at	timestamp			Yes	NULL			Change Drop More

Departments

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐ 2	name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 3	description	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 4	manager_id 🔑	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐ 5	status	enum('active', 'inactive')	utf8mb4_unicode_ci		No	active			Change Drop More
☐ 6	created_at	timestamp			Yes	NULL			Change Drop More
☐ 7	updated_at	timestamp			Yes	NULL			Change Drop More

Employees

☐ 1	id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐ 2	user_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐ 3	department_id 🔑	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐ 4	first_name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 5	last_name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 6	email 🔑	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 7	phone	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 8	address	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 9	position	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 10	hire_date	date			No	None			Change Drop More
☐ 11	employment_status	enum('active', 'inactive', 'terminated')	utf8mb4_unicode_ci		No	active			Change Drop More
☐ 12	created_at	timestamp			Yes	NULL			Change Drop More
☐ 13	updated_at	timestamp			Yes	NULL			Change Drop More

Attendances

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐ 2	employee_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐ 3	attendance_date	date			No	None			Change Drop More
☐ 4	time_in	time			Yes	NULL			Change Drop More
☐ 5	time_out	time			Yes	NULL			Change Drop More
☐ 6	status	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 7	remarks	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 8	created_at	timestamp			Yes	NULL			Change Drop More
☐ 9	updated_at	timestamp			Yes	NULL			Change Drop More

Leave Request

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
2	employee_id	bigint(20)		UNSIGNED	No	None			Change Drop More
3	leave_type	enum('Vacation Leave', 'Sick Leave', 'Emergency Le...')	utf8mb4_unicode_ci		No	None			Change Drop More
4	start_date	date			No	None			Change Drop More
5	end_date	date			No	None			Change Drop More
6	reason	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
7	status	enum('pending', 'approved', 'rejected')	utf8mb4_unicode_ci		No	pending			Change Drop More
8	approved_by	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
9	approved_at	timestamp			Yes	NULL			Change Drop More
10	created_at	timestamp			Yes	NULL			Change Drop More
11	updated_at	timestamp			Yes	NULL			Change Drop More

X. ADVANCED DATA FEATURES

Triggers

1. **trg_attendance_no_duplicate_timein (ATTENDANCE TABLE)** - The purpose of this trigger is to prevent duplicate time-in entries for a single attendance record. It ensures that an employee cannot overwrite or insert multiple time-in values for the same attendance day. This improves data integrity by protecting the system from duplicate submissions caused by repeated requests or accidental user actions, ensuring that each attendance record has only one valid time-in entry.

Name	Time	Event	
☐ trg_attendance_no_duplicate_timein	BEFORE	UPDATE	Edit Export Drop

2. **trg_attendance_set_remarks (ATTENDANCE TABLE)** - The purpose of this trigger is to automatically assign the correct attendance remark such as Undertime, Complete, or Overtime based on the employee's total rendered working hours. This ensures that attendance records remain accurate even when updates are made through different parts of the system such as HR manual editing or employee time-out actions. Instead of relying on application-level computation, this trigger guarantees that the remarks are always updated directly in the database whenever a time-out value is recorded or modified.

☐ trg_attendance_set_remarks	BEFORE	UPDATE	Edit Export Drop
--	--------	--------	--------------------

3. **trg_attendance_validate_timeout (ATTENDANCE TABLE)** - The purpose of this trigger is to ensure that the time-out value is always greater than the time-in value. This prevents invalid attendance data such as negative working hours or incorrect time

calculations. By enforcing this rule at the database level, the system ensures that all attendance records maintain logical and accurate time sequences even when edited manually by HR or administrators.

☐ **trg_attendance_validate_timeout** BEFORE UPDATE  Edit  Export  Drop

Views

1. **vw_monthly_attendance_summary** - The purpose of this view is to provide a monthly summary of employee attendance including total present days, absences, leave days, overtime, and total rendered working hours. This view simplifies monthly reporting and payroll preparation by consolidating attendance data into a single structured output. It eliminates the need for repeated monthly aggregation queries across multiple tables.

employee_id	full_name	department	year	month	present_days	late_days	absent_days	on_leave_days	overtime_days	undertime_days	total_hours_worked
7	Jisung Park	Information Technology	2026	4	0	0	4	0	0	0	0.00
8	Chenle Zhong	Human Resources	2026	4	1	3	3	1	1	0	8.01
9	Yuta Nakamoto	Human Resources	2026	4	1	3	3	0	0	1	7.78
10	Jaemin Naa	Human Resources	2026	4	1	1	4	0	NULL	NULL	0.00

2. **vw_employee_attendance_summary** - The purpose of this view is for the attendance summary reports of the system that makes it easy to see who is present, absent on a given day along with their department information. This aids in terms of making the query easy instead of querying multiple tables separately, this view provides a ready to use summary in one view. It is because this view combines three tables which are the attendances, employees, and departments to display a complete attendance record for each employee including their full name and their department.

employee_id	full_name	position	department	total_records	present_count	late_count	absent_count	on_leave_count	overtime_count	undertime_count	avg_hours_worked
5	Renjun Huang	Hr	NULL	0	NULL	NULL	NULL	NULL	NULL	NULL	NULL
7	Jisung Park	Head	Information Technology	4	0	0	4	0	0	0	NULL
8	Chenle Zhong	Staff	Human Resources	8	1	3	3	1	1	0	8.01
9	Yuta Nakamoto	Head	Human Resources	7	1	3	3	0	0	1	7.78
10	Jaemin Naa	Head	Human Resources	6	1	1	4	0	NULL	NULL	NULL
12	Mark Lee	Manager	Human Resources	0	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. **vw_department_attendance_today** - The purpose of this view is to display the current day's attendance records of employees together with their department information and computed working hours. It helps simplify daily monitoring of employee attendance per

department by combining relevant data into a single query result instead of retrieving it from multiple tables separately. This makes it easier for managers to monitor attendance status in real time.

department_id	department_name	employee_name	position	attendance_date	time_in	time_out	status	remarks	hours_worked
2	Information Technology	Jisung Park	Head	NULL	NULL	NULL	NULL	NULL	NULL
1	Human Resources	Chenle Zhong	Staff	2026-04-30	NULL	NULL	On Leave	Approved leave	NULL
1	Human Resources	Yuta Nakamoto	Head	2026-04-30	07:58:43	NULL	Present	NULL	NULL
1	Human Resources	Jaemin Naa	Head	2026-04-30	07:58:08	NULL	Present	NULL	NULL
1	Human Resources	Mark Lee	Manager	NULL	NULL	NULL	NULL	NULL	NULL

XI. SYSTEM FEATURES AND FUNCTIONALITIES

Authentication Module

This module is responsible for managing the user access to the system. This ensures that only authorized users can access and can log in in the system and also access the system features based on their assigned roles. This module is important in the system because it maintains the security and protects sensitive information.

Features:

- Allow users to log in to the system
- Allow users to log out securely
- Ensure password security through hashing
- Implement role-based access using middleware
- Enable users to manage their profile information

Dashboard Module

This module provides an overview for key HR data and system activities. This allows users, especially HR staff and managers to quickly monitor important records or information and make informed decisions based on the real time summaries.

Features:

- Display total number of employees

- Display total number of departments
- Show employees present for the current day
- Display pending leave requests
- Show recent HR-related activities

Employee Module

This Module handles the storage and maintenance of employee information. This serves as the central component of the system, allowing HR staff to manage employee records efficiently and keep all relevant information updated and organized.

Features:

- Add new employee records
- Edit existing employee information
- Delete employee records
- View detailed employee profiles
- Search and filter employee records
- Assign employees to departments

Department Module

This Module is responsible for organizing employees into different departments within the organization. It helps maintain a clear structure and allows better management of employees based on their assigned roles and departments.

Features:

- Add new departments
- Edit department details
- Delete departments
- View department list
- Assign a manager to each department

Attendance Module

The Attendance Management Module tracks the daily attendance and work status of employees. It helps HR staff and managers monitor employee punctuality and attendance patterns, which are important for evaluating performance and ensuring accountability.

Features:

- Record employee time-in
- Record employee time-out
- View attendance history
- Assign attendance status (Present, Late, Absent, On Leave)
- Filter attendance by date or employee

Leave Request Module

This module manages the process of employee leave applications and approvals. This allows employees to submit leave requests and enables managers or HR staff to review, approve, or reject these requests, ensuring a structured workflow.

Features:

- Submit leave requests
- Approve leave requests
- Reject leave requests
- View leave request history
- Track leave request status

Reports Modules

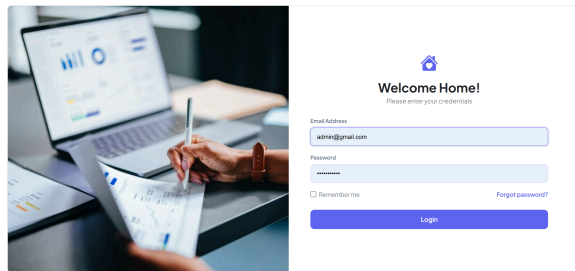
In this module it generates summarized information that helps HR staff and management monitor employee data and system activity. These reports support decision-making and provide insights into workforce trends and performance.

Features:

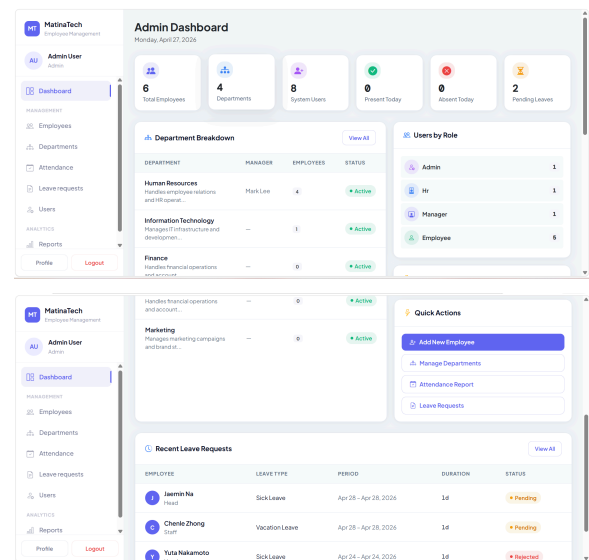
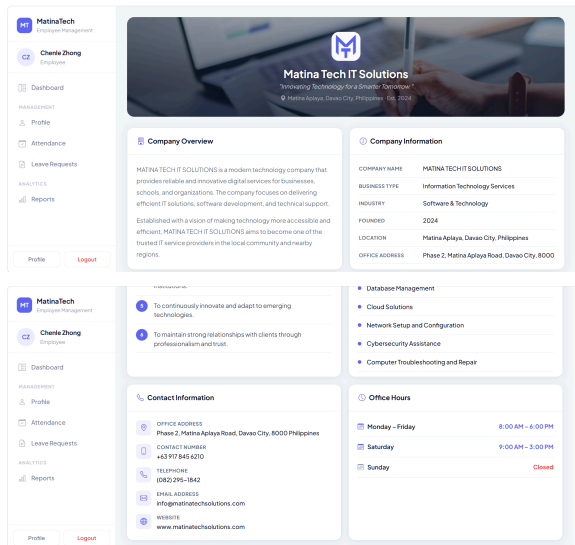
- Generate employee list reports
- Generate attendance summary reports
- Generate leave request reports
- Generate department-based reports
- Generate monthly HR activity reports

XII. PROTOTYPE / USER INTERFACE

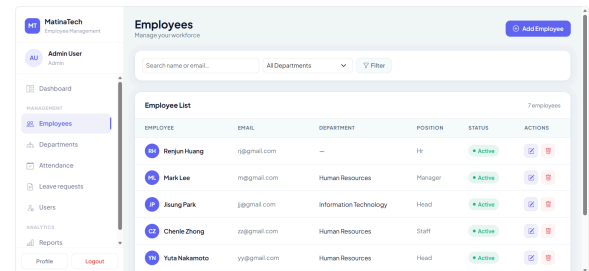
LOGIN PAGE



INSTITUTIONAL PAGE



ADMIN PAGE - EMPLOYEES/INDEX



ADMIN PAGE - DASHBOARD

ADMIN PAGE - EMPLOYEE/CREATE

MatinaTech

Employee Management

Admin User

Admin

Dashboard

MANAGEMENT

Employees

Departments

Attendance

Leave requests

Users

ANALYTICS

Reports

Profile

Logout

Create New Employee

Fill in the details below to add a new employee.

Employee Information

FIRST NAME

e.g. Juan

LAST NAME

e.g. dela Cruz

EMAIL

juand@matinatech.com

PHONE

+63 900 XXX XXXX

ADDRESS

Street, Barangay, City

DEPARTMENT

Select a Department

POSITION

e.g. Software Engineer

HIRE DATE

mm/dd/yyyy

EMPLOYMENT STATUS

Active

Cancel

Create Employee

ADMIN PAGE - DEPARTMENT/CREATE

MatinaTech

Employee Management

Admin User

Admin

Dashboard

MANAGEMENT

Employees

Departments

Attendance

Leave requests

Users

ANALYTICS

Reports

Profile

Logout

Add Department

Fill in the details below to create a new department.

Department Information

DEPARTMENT NAME

e.g. Engineering

MANAGER

Select Manager

DESCRIPTION

Brief description of this department's responsibilities...

STATUS

Active

Cancel

Create Department

ADMIN PAGE - DEPARTMENT/EDIT

MatinaTech

Employee Management

Admin User

Admin

Dashboard

MANAGEMENT

Employees

Departments

Attendance

Leave requests

Users

ANALYTICS

Reports

Profile

Logout

Edit Department

Update the department's details.

Department Information

DEPARTMENT NAME

Human Resources

MANAGER

Mark Lee

DESCRIPTION

Handles employee relations and HR operations

STATUS

Active

Cancel

Save Changes

ADMIN PAGE - EMPLOYEE/EDIT

MatinaTech

Employee Management

Admin User

Admin

Dashboard

MANAGEMENT

Employees

Departments

Attendance

Leave requests

Users

ANALYTICS

Reports

Profile

Logout

Edit Employee

Update the employee's personal and employment details.

Personal Information

FIRST NAME

Benjun

LAST NAME

Huang

EMAIL

rb@gmail.com

PHONE

+63 900 XXX XXXX

ADDRESS

Street, Barangay, City

EMPLOYMENT INFORMATION

POSITION

HR

DEPARTMENT

Select Department

HIRE DATE

04/12/2026

EMPLOYMENT STATUS

Active

Cancel

Save Changes

ADMIN PAGE - ATTENDANCE/SHOW

MatinaTech

Employee Management

Admin User

Admin

Dashboard

MANAGEMENT

Employees

Departments

Attendance

Leave requests

Users

ANALYTICS

Reports

Profile

Logout

Attendance Reports

Filter and manage all attendance records.

DATE

mm/dd/yyyy

EMPLOYEE

Search name...

STATUS

All Status

Filter

All Records

18 records

EMPLOYEE	DEPARTMENT	DAY / DATE	TIME IN	TIME OUT	STATUS	HOURS	REMARKS	ACTIONS
Jaemin Na	Human Resources	Monday Apr 27, 2026	04:04 PM	06:47 PM	Late	2.72 hrs	Undertime	🗕 🗑
Yuta Nakamoto	Human Resources	Monday Apr 27, 2026	04:04 PM	06:04 PM	Late	4.01 hrs	Undertime	🗕 🗑
Chenle Zhong	Human Resources	Monday Apr 27, 2026	04:02 PM	--	Late	0 hrs	--	🗕 🗑

File Explorer

ADMIN PAGE - ATTENDANCE/EDIT

MatinaTech

Employee Management

Admin User

Admin

Dashboard

MANAGEMENT

Employees

Departments

Attendance

Leave requests

Users

ANALYTICS

Reports

Profile

Logout

Edit Attendance Record

Update the attendance details for the record.

Attendance Details

STATUS

Late

TIME IN

04:04 PM

TIME OUT

06:47 PM

REMARKS

Undertime

Cancel

Save Changes

ADMIN PAGE - DEPARTMENT/INDEX

MatinaTech

Employee Management

Admin User

Admin

Dashboard

MANAGEMENT

Employees

Departments

Attendance

Leave requests

Users

ANALYTICS

Reports

Profile

Logout

Departments

Manage your organization's departments.

Add Department

Search by name...

All Status

Filter

Department List

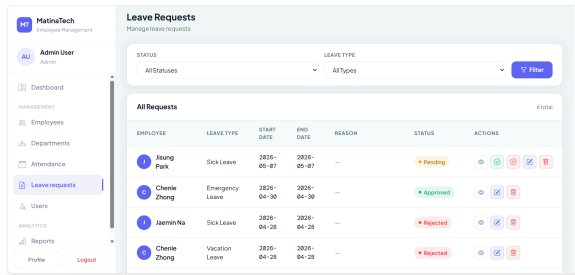
4 departments

DEPARTMENT	MANAGER	STATUS	ACTIONS
Human Resources	Mark Lee	Active	🗕 🗑
Information Technology	--	Active	🗕 🗑
Finance	--	Active	🗕 🗑
Marketing	--	Active	🗕 🗑

ADMIN PAGE - LEAVE

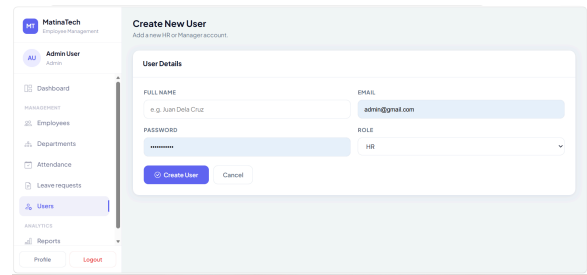
16

REQUESTS/INDEX



EMPLOYEE	LEAVE TYPE	START DATE	END DATE	REASON	STATUS	ACTIONS
Jaung Park	Sick Leave	2026-01-27	2026-02-27		Pending	[Icon] [Icon] [Icon] [Icon]
Cherie Zhong	Emergency Leave	2026-04-30	2026-04-30		Approved	[Icon] [Icon] [Icon]
Jaemin Na	Sick Leave	2026-04-28	2026-04-28		Rejected	[Icon] [Icon] [Icon]
Cherie Zhong	Vacation Leave	2026-04-28	2026-04-28		Rejected	[Icon] [Icon] [Icon]

ADMIN PAGE - USERS/CREATE



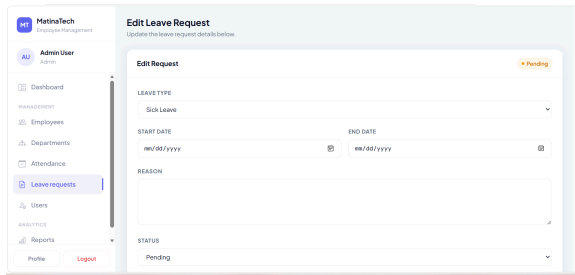
Create New User
Add a new HR or Manager account.

User Details

FULL NAME: e.g. Jon Doe Cruz
EMAIL: admin@gmail.com
PASSWORD: [Masked]
ROLE: HR

[Create User] [Cancel]

ADMIN PAGE - LEAVE REQUESTS/EDIT



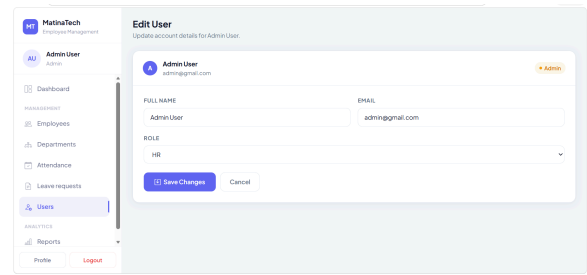
Edit Leave Request
Update the leave request details below.

Edit Request [Pending]

LEAVE TYPE: Sick Leave
START DATE: mm/dd/yyyy
END DATE: mm/dd/yyyy
REASON: [Text Area]
STATUS: Pending

[Save Changes] [Cancel]

ADMIN PAGE - USERS/EDIT

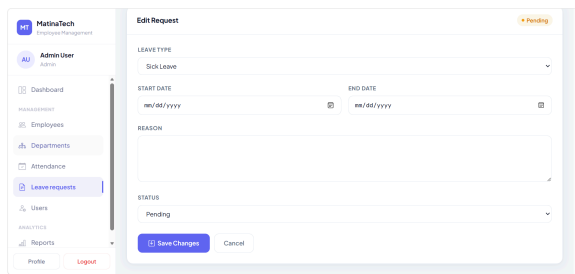


Edit User
Update account details for Admin User.

Admin User [Admin]

FULL NAME: Admin User
EMAIL: admin@gmail.com
ROLE: HR

[Save Changes] [Cancel]



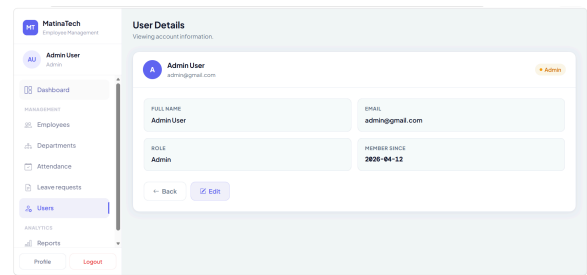
Edit Leave Request
Update the leave request details below.

Edit Request [Pending]

LEAVE TYPE: Sick Leave
START DATE: mm/dd/yyyy
END DATE: mm/dd/yyyy
REASON: [Text Area]
STATUS: Pending

[Save Changes] [Cancel]

ADMIN PAGE - USERS/VIEW



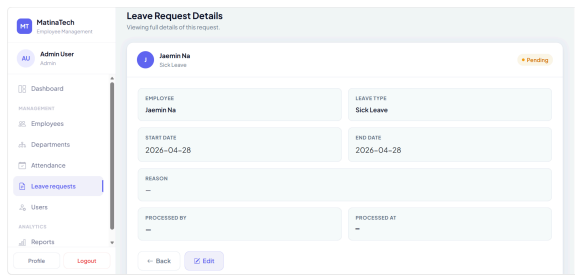
User Details
Viewing account information.

Admin User [Admin]

FULL NAME: Admin User
EMAIL: admin@gmail.com
ROLE: Admin
HIRED SINCE: 2026-04-12

[Back] [Edit]

ADMIN PAGE - LEAVE REQUEST/VIEW



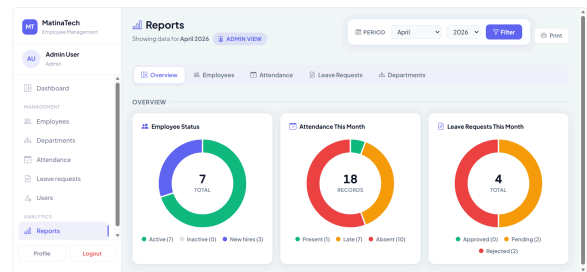
Leave Request Details
Viewing full details of this request.

Jaemin Na [Pending]

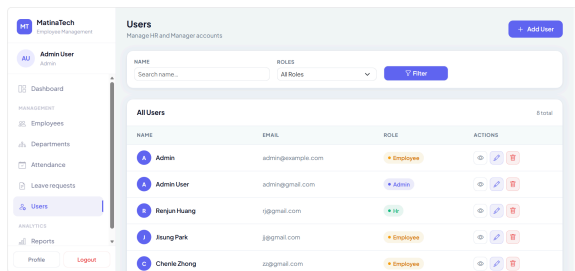
EMPLOYEE: Jaemin Na
LEAVE TYPE: Sick Leave
START DATE: 2026-04-28
END DATE: 2026-04-28
REASON: [Text Area]
PROCESSED BY: [Text Area]
PROCESSED AT: [Text Area]

[Back] [Edit]

ADMIN PAGE - REPORTS



ADMIN PAGE - USERS/INDEX

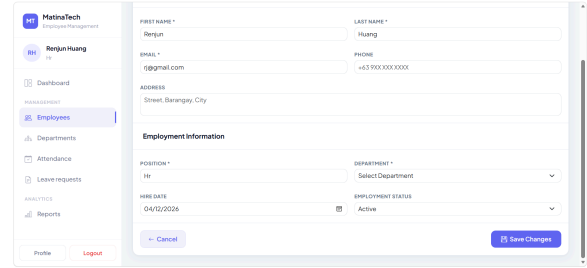
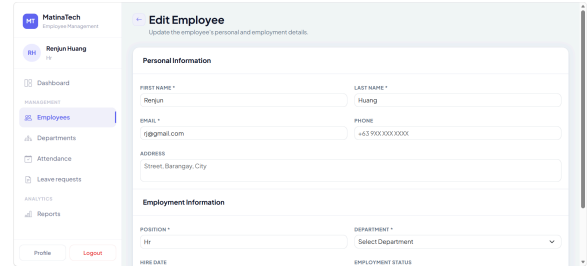
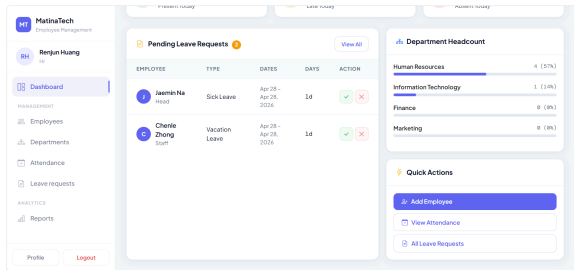
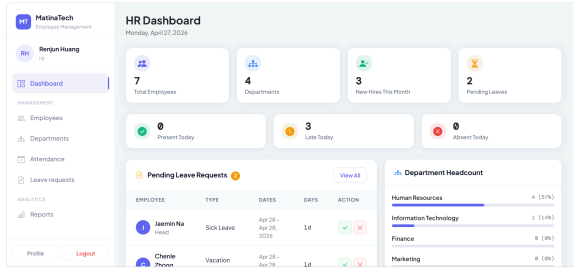


Users
Manage HR and Manager accounts.

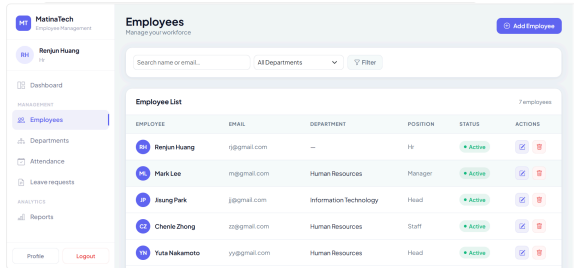
NAME: [Search Name] | ROLE: All Roles [Filter]

NAME	EMAIL	ROLE	ACTIONS
Admin	admin@example.com	Employee	[Icon] [Icon] [Icon]
Admin User	admin@gmail.com	Admin	[Icon] [Icon] [Icon]
Renjun Huang	rg@gmail.com	HR	[Icon] [Icon] [Icon]
Jaung Park	j@gmail.com	Employee	[Icon] [Icon] [Icon]
Cherie Zhong	z@gmail.com	Employee	[Icon] [Icon] [Icon]

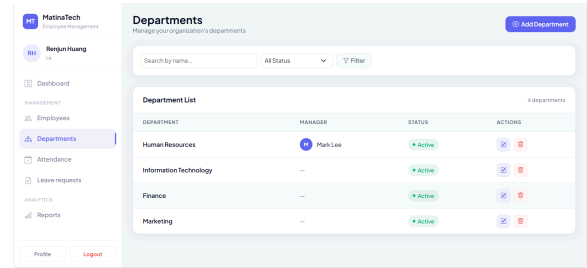
HR PAGE - DASHBOARD



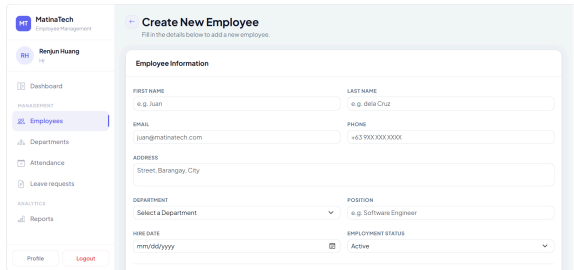
HR PAGE - EMPLOYEE/INDEX



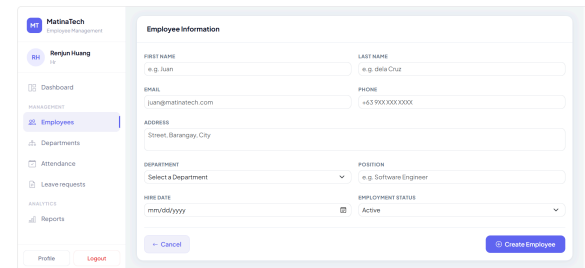
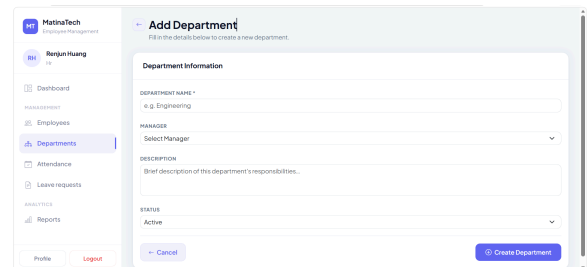
HR PAGE - DEPARTMENTS/INDEX



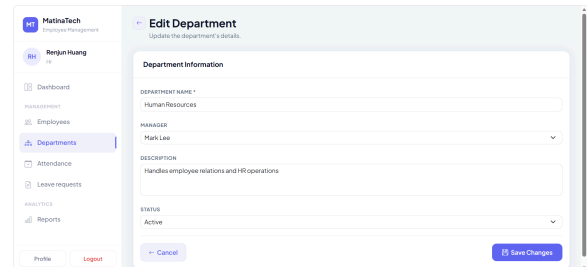
HR PAGE - EMPLOYEE/CREATE



HR PAGE - DEPARTMENTS/CREATE

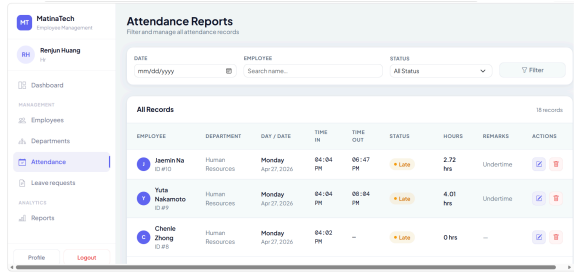


HR PAGE - DEPARTMENTS/EDIT

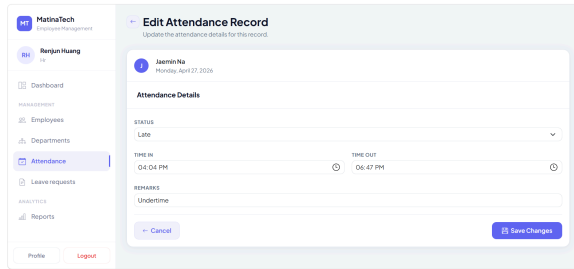


HR PAGE - EMPLOYEE/EDIT

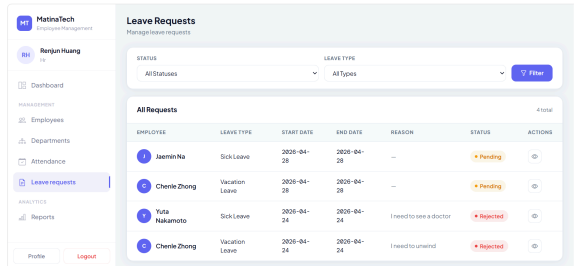
HR PAGE - ATTENDANCE/INDEX



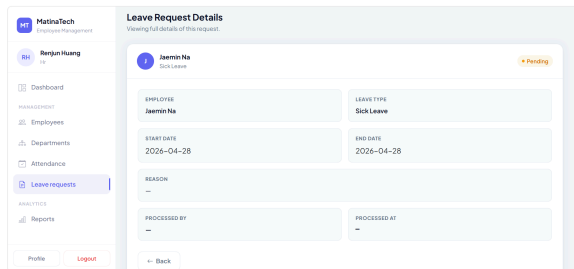
HR PAGE - ATTENDANCE/EDIT



HR PAGE - LEAVE REQUESTS/INDEX



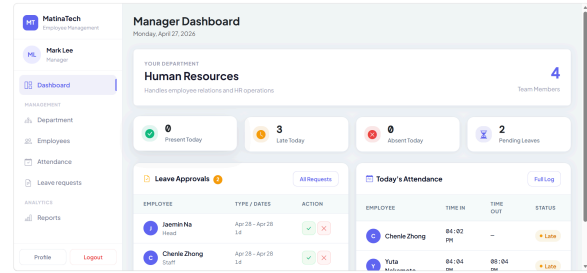
HR PAGE - LEAVE REQUESTS/REVIEW



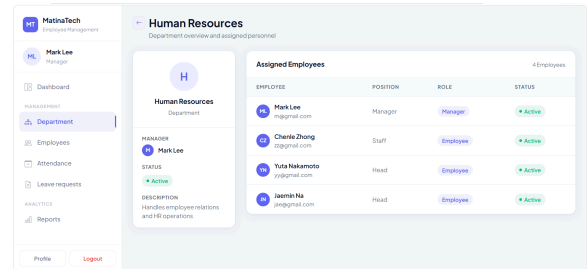
HR PAGE - REPORTS



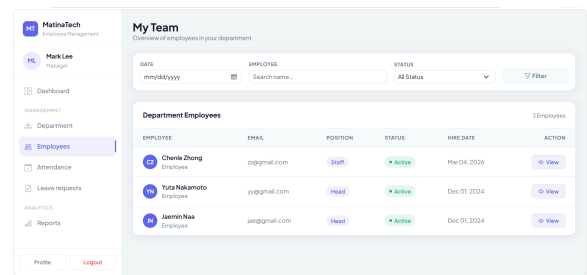
MANAGER PAGE - DASHBOARD



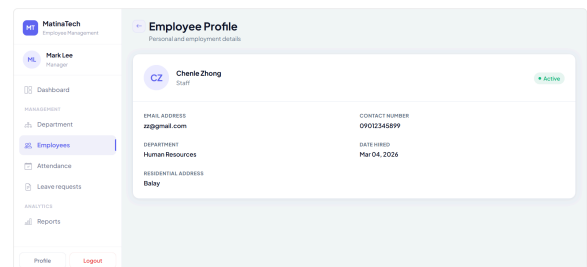
MANAGER PAGE - DEPARTMENT/VIEW



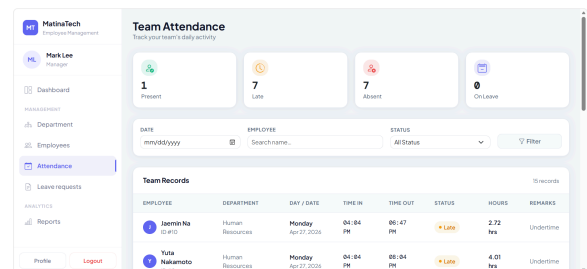
MANAGER PAGE - EMPLOYEE/VIEW



MANAGER PAGE - EMPLOYEE/VIEW



MANAGER PAGE - ATTENDANCE/VIEW



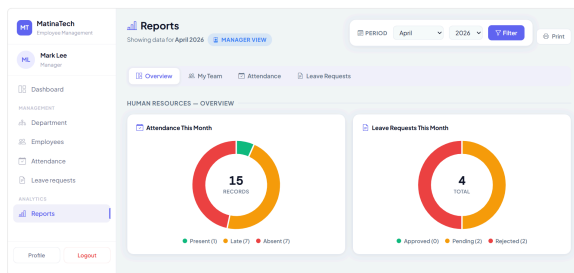
MANAGER PAGE - LEAVE REQUESTS/INDEX

EMPLOYEE	LEAVE TYPE	START DATE	END DATE	REASON	STATUS	ACTIONS
Jaemin Na	Sick Leave	2026-04-28	2026-04-28	—	Pending	
Charlie Zhong	Vacation Leave	2026-04-28	2026-04-28	—	Pending	
Yuta Nakamoto	Sick Leave	2026-04-24	2026-04-24	I need to see a doctor	Rejected	
Charlie Zhong	Vacation Leave	2026-04-24	2026-04-24	I need to unwind	Rejected	

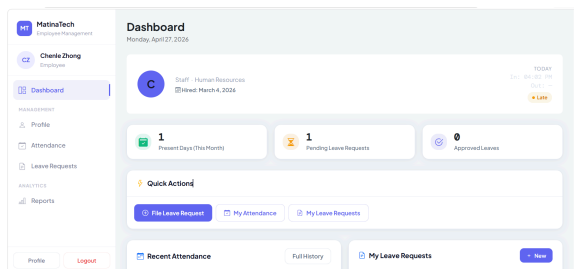
MANAGER PAGE - LEAVE REQUESTS/VIEW

EMPLOYEE	LEAVE TYPE	START DATE	END DATE	REASON	PROCESSED BY	PROCESSED AT
Jaemin Na	Sick Leave	2026-04-28	2026-04-28	—	—	—

HR PAGE - REPORTS



EMPLOYEE PAGE - DASHBOARD



DATE	TIME IN	TIME OUT	STATUS	TYPE	DATES	STATUS
Apr 27, 2026	04:02 PM	—	Late	Vacation Leave	Apr 28 - Apr 28, 2026	Pending
Apr 26, 2026	—	—	Absent	Vacation Leave	Apr 24 - Apr 24, 2026	Rejected
Apr 23, 2026	12:23 PM	—	Late	—	—	—
Apr 23, 2026	—	—	Absent	—	—	—
Apr 17, 2026	08:11 AM	04:11 PM	Present	—	—	—
Apr 14, 2026	05:01 PM	—	Late	—	—	—

EMPLOYEE PAGE - EMPLOYEE/INFORMATION ONLY

Employee Profile

Personal Information

First Name: Charlie, Last Name: Zhong, Email Address: cz@gmail.com, Residential Address: Bala, Contact Number: 0902345678

Employment Information

Job Title: Staff, Department: Human Resources, Date Hired: March 04, 2026, Employment Status: Active

EMPLOYEE PAGE - ATTENDANCE

My Attendance

Current Time: 9:56:18 PM

Status: Late

Attendance History

DAY	DATE	TIME IN	TIME OUT	STATUS	HOURS	REMARKS
Monday	Apr 27, 2026	04:02 PM	09:05 PM	Late	5.88 hrs	Undertime
Sunday	Apr 26, 2026	—	—	Absent	0 hrs	—
Thursday	Apr 23, 2026	12:23 PM	—	Late	0 hrs	—
Wednesday	Apr 22, 2026	—	—	Absent	0 hrs	—

EMPLOYEE PAGE - LEAVE REQUESTS/INDEX

EMPLOYEE	LEAVE TYPE	START DATE	END DATE	REASON	STATUS	ACTIONS
Yuta Nakamoto	Sick Leave	2026-04-24	2026-04-24	I need to see a doctor	Rejected	

EMPLOYEE PAGE - LEAVE REQUESTS/CREATE

MatinaTech
Employee Management

Apply for Leave
Please provide details below to submit a leave request.

Leave Request Form

LEAVE TYPE
Select Leave Type

START DATE
mm/dd/yyyy

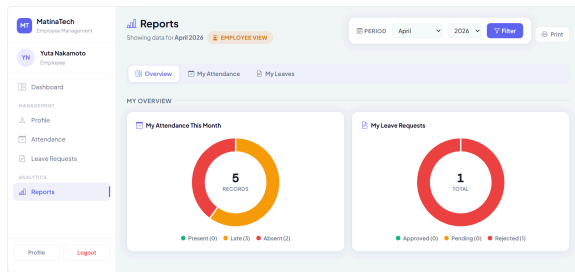
END DATE
mm/dd/yyyy

REASON (optional)
Briefly describe your reason...

[Submit Request](#) [Cancel](#)

Profile [Logout](#)

EMPLOYEE PAGE - REPORTS



XIII. TESTING AND RESULTS

LOG-IN MODULE

TEST CASE	DESCRIPTION	EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
TC.1.1	User logs in with valid email and password	User is authenticated and redirected to dashboard based on their role	User was successfully authenticated and redirected to the correct role-based dashboard	PASSED
TC.1.2	User logs in with invalid credentials (wrong email/password)	Login fails and shows "These credentials do not match our records" error	Login was rejected and error message was displayed	PASSED
TC.1.3	User submits empty login form	System displays validation errors for required fields	Validation errors were displayed for email field "Please fill-out this field"	PASSED
TC.1.4	Deactivated or terminated user attempts to log in	User is redirected to Account Deactivated page	User was redirected to Account Deactivated page	PASSED

EMPLOYEE MODULE

TEST CASE	DESCRIPTION	EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
-----------	-------------	-----------------	---------------	--------

T.C.2.1	Employee is logged in AND employment_status is 'inactive' or 'terminated'	Employee is redirected to the deactivated page and Protected modules remain inaccessible	“Your account has been deactivated or terminated. Please contact HR or your administrator for assistance.”	PASSED
T.C.2.2	HR/Admin submits valid employee information.	The system should successfully insert a new employee record into the database along with its corresponding user account record.	The system successfully insert a new employee record into the database along with its corresponding user account record.	PASSED
T.C.2.2	HR/Admin submits employee information using an existing email address.	The system should reject the transaction and prevent duplicate email records from being stored.	“Please fix the following errors: The email has already been taken”. The database didn't save as it has UNIQUE constraints in the email column.	PASSED
T.C.2.3	HR/Admin modifies employee information.	The system should successfully update the employee record in the database.	The system successfully update the employee record in the database.	PASSED
T.C.2.4	HR/Admin deletes an employee record.	The system should remove the employee record and its related records according to cascade delete constraints.	The system removed the employee record and its related records according to cascade delete constraints.	PASSED
TC.2.5	HR/Admin changes employee status to Terminated.	The employee status field should be updated and access restrictions should apply.	The employee status field is updated and access restrictions should apply.	PASSED
TC.2.5	HR/Admin changes employee status back to Active.	The employee record should be updated successfully and system access restored.	The employee record is updated successfully and system access restored.	PASSED

TC.2.6	The manager views employees under their department.	The system should retrieve only employee records associated with the manager's department.	The system retrieves only employee records associated with the manager's department.	PASSED
TC.2.7	An employee record exists without an assigned department.	The system should handle null department relationships without database or retrieval errors.	The system handled null department relationships without database or retrieval errors.	PASSED
TC.2.8	HR/Admin deletes an employee assigned as department manager.	The employee record should be deleted while the related department manager reference becomes NULL.	The employee record is deleted while the related department manager reference becomes NULL.	PASSED

DEPARTMENT MODULE

TEST CASE	DESCRIPTION	EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
TC.3.1	Admin creates a department with valid details and assigns a manager	The department record is saved in the database and displayed in the department list with the assigned manager	The department was successfully saved and displayed with the assigned manager	PASSED
TC.3.2	Admin creates a department without assigning a manager	The department record is saved successfully with a NULL manager field	The department was successfully saved with no manager assigned (NULL value)	PASSED
TC.3.3	Admin updates an existing department and changes the assigned manager	The department record is updated and reflects the new manager assignment	The department was successfully updated and the new manager was reflected in the list	PASSED
TC.3.4	Admin deletes a department with no associated employees	The department record is removed from the database and no longer appears in the list	The department was successfully deleted and removed from the list	PASSED
TC.3.5	Admin deletes a department that has associated employees	The department is deleted and related employees have their	The department was deleted and employees' department_id was set	PASSED

		department_id set to NULL (based on cascade rules)	to NULL	
TC.3.6	Admin sets department status to inactive	The department status is updated to Inactive and reflected in the department list	The department status was successfully updated to Inactive	PASSED
TC.3.7	Manager views their assigned department	The system displays only the manager's assigned department record	Only the assigned department was displayed to the manager	PASSED
TC.3.8	Admin views all departments	All department records are displayed with correct manager and status information	All departments were displayed with correct data	PASSED

ATTENDANCE MODULE

TEST CASE	DESCRIPTION	EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
TC.4.0	The employee did not time in before 11:59 PM.	The employee's time in/out button is already in a completed state. The status should be marked as absent.	The employee's time in/out button is already in a completed state. The status should be marked as absent.	PASSED
T.C.4.1	The employee status is on leave	The employee's time in/out button is already in a completed state.	The employee's time in/out button is already in a completed state.	PASSED
T.C.4.2	Employee times in before 8:15 AM	Status should be marked as Present	Status is marked as Present	PASSED
T.C.4.3	Employee times in after 8:15 AM	Status should be marked as Late	Status is marked as Late	PASSED
T.C.4.4	Employee tries to time in twice on the same day	The system does not allow double time in, the system automatically sets the buttons to time out after timing in.	Second time-in attempt is blocked	PASSED
T.C.4.5	Employee tries to time in on a day already marked as Absent	System should block time-in action	Time-in is blocked	PASSED

T.C.4.6	Employee times out successfully	System assigns remarks based on total hours: Complete, Undertime, or Overtime	Remarks are correctly assigned based on hours	PASSED
T.C.4.7	Admin edits an existing attendance record	Admin should be able to update attendance details successfully	Attendance record is updated	PASSED
T.C.4.8	Employee, admin, hr, manager filters records by date, employee, and status	System should display only matching records	Filtered results are displayed correctly	PASSED
T.C.4.9	The employee successfully times in but does not perform a time-out for the day.	Total hours should be 0 hours	Hours remain 0 and remarks are blank.	PASSED
T.C.10	Admin deleted an attendance records	Records should be removed from the system and the database.	Record is deleted successfully	PASSED

LEAVE REQUESTS MODULE

TEST CASE	DESCRIPTION	EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
TC.5.0	The pending request has passed the start date.	The system will automatically reject the request.	The system automatically rejected the request.	PASSED
TC.5.1	The employee submits a leave request with a start date set for tomorrow.	The leave request should still be displayed in the table.	The leave request is displayed in the table.	PASSED
TC.5.2	The employee submits a leave request with a start date in the past.	The employee cannot press the past dates.	The employee cannot press the past dates.	PASSED

TC.5.3	An employee who has a pending leave request, requested another leave request with different dates.	The employee can still request a leave request with different dates.	The employee can still request a leave request with different dates.	PASSED
TC.5.4	A leave request has been approved by the admin.	The request should be displayed with “Approved” status.	The request should be displayed with “Approved” status.	PASSED
T.C.5.5	Employee views their own leave requests	The system should display only the logged-in employee’s leave requests.	The system displays only the logged-in employee’s leave requests.	PASSED
T.C.5.6	Employee submits a leave with overlapping dates on an existing pending request	The employee cannot submit another leave request that overlaps with a selected dates	“You already have a pending or approved leave that overlaps with the selected dates. ”	PASSED
T.C.5.7	Employee submits a leave with overlapping dates on an already approved request	The employee cannot submit another leave request that overlaps with a selected dates	“You already have a pending or approved leave that overlaps with the selected dates. ”	PASSED
T.C.5.8	The employee submits a leave request where end_date is before start_date	The system should reject the request and display an error message indicating that the end date cannot be earlier than the start date.	The system displays an error message “The end date field must be a date after or equal to the start date.”	PASSED
T.C.5.9	Submit leave during a date already marked as Absent or Present	The system should prevent the leave request from being submitted for that date and display an error message indicating that leave cannot be filed for an already recorded attendance date.	“ You already have an attendance record on one or more of the selected dates. ”	PASSED

USER MODULE

TEST CASE	DESCRIPTION	EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
TC.6.1	Admin creates a new	User account is created	User was created	PASSED

	HR or Manager user with valid detail1	successfully and linked to an employee record	successfully and employee record was auto-linked	
TC.6.2	Admin creates a user with duplicate email	System displays validation error and prevents creation	“The email has already been taken.”	PASSED
TC.6.3	Admin creates a user with invalid password (less than 8 characters)	System rejects input and shows password validation error	Password validation error was displayed	PASSED
TC.6.4	Admin updates user with duplicate email or invalid role	System displays validation error and prevents update	“The email has already been taken.”	PASSED

XIV. CONCLUSION AND RECOMMENDATION

CONCLUSION

The development of the system employee management system for Matinatech IT Solutions, Inc. has successfully accomplished its primary objectives of providing Matinatech IT Solutions a centralized and web-based platform to manage its employee-related processes. This study addresses the challenges associated with Matinatech's previous manual employee-related processes by building essential human resource operations such as employee and department record management, attendance monitoring, department organization, and also leave request processing. Through this implementation of these functionalities into a single platform, the system improved the operational efficiency and reduced data redundancy and also minimized the occurrence of human errors.

Additionally, this system implemented a role-based access management to ensure that users, including admins, HR, managers, and employees, could only access the features that are relevant and created for them and for their designated responsibilities. The integration of database functionalities such as CRUD operations, triggers, and views contributed highly to maintaining data integrity and improved record management, simplifying attendance computation and the reporting process. These features enhanced both the security and reliability of the system.

In conclusion, based on the conducted testing and evaluation procedures. All of the major modules of the system functioned according to the specified rules and requirements with test cases yielding successful results. The finding indicates that the development of the system is reliable and functional and capable of supporting the maintenance tech solution operational needs. Overall the employee management system serves as an effective solution for improving the workflow efficiency and enhances the data management and supporting organizational productivity within the Matinatech IT solutions.

RECOMMENDATIONS

Although the developed Employee Management System successfully achieved its intended objectives, several enhancements are recommended to further improve its functionality, efficiency, security, and user experience in future developments.

- Integration of Payroll Management Module, Future development may include payroll-related functionalities such as salary computation, overtime calculation, deduction management, tax computation, and payslip generation to create a more comprehensive Human Resource Information System (HRIS).
- Biometric or RFID Attendance Integration, The system may be integrated with biometric devices or RFID-based attendance monitoring to improve attendance accuracy, reduce manual input errors, and prevent unauthorized attendance entries.
- Real-Time Notification System, Implementing email or in-application notifications for leave approvals, attendance concerns, announcements, and account updates may improve communication and responsiveness among users.
- User Interface and User Experience Enhancements, Further improvements in navigation, responsive design, accessibility, and visual consistency may enhance usability and overall user satisfaction.
- Audit Logs and Activity Tracking, Adding audit logs and activity monitoring features would improve accountability and transparency by recording user actions such as edits, deletions, approvals, and login activities within the system.

- Database Backup and Disaster Recovery Mechanism, The implementation of automated backup systems and disaster recovery solutions is recommended to ensure data security, reliability, and recovery during unexpected system failures.

XV. REFERENCES

[1] Bajpayi, R., Sharma, M. L., & Tripathi, K. C. (2020). Employee management system. *International Journal for Modern Trends in Science and Technology*, 6(12), 225–234. <https://doi.org/10.46501/IJMTST061242>

[2] Engr. Mosud Y. Olumoye, “The Development of computer-based staff management system”, EIJSST Vol-2, pages 41-51, Nov-13.

[3] Prety Diawati, Vip Paramarta , Djoko Pitoyo , Tomy Fitrio, Sri Wiyati Mahrani, “Challenges of Implementing an Employee Management System for Improving Workplace Management Effectiveness ”, *Journal of Environmental Treatment Techniques* , Special Issue on Environment, Management and Economy, Pages: 1200-1203,2019.

[4] Jahmani, A., Jawabreh, O., Fahmawee, E. A. A. L., Almasarweh, M., & Ali, B. J. (2023). *The impact of employee management on organizational performance in Dubai's five-star hotel sector*. Journal of Statistics Applications & Probability, 12(2), 395–404.

[5] Farzana, M. A. (2023). *EmploEdge: An employee management system* (Internship/project report, Independent University, Bangladesh).

[6] Ning, K. C., Ahmad, J., Zakaria, N. H., & Yuen, S. C. K. (2024). *Employee management system for enterprise companies*. International Journal of Innovative Computing, 14(1), 57–61. <https://doi.org/10.11113/ijic.v14n1.467>